Psychometric Utility Theory

A Formal Framework for the Dynamics of
Human Motivation, Betrayal, and Collapse

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"Everyone sees what you appear to be, few experience what you really are."
— Niccolò Machiavelli, *The Prince*, Ch. XVIII

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Abstract

Classical utility theory models agents as rational optimizers over preferences. This assumption fails catastrophically when applied to real human behavior—where betrayal, self-delusion, identity substitution, and emotional collapse are not exceptions but *structural features* of the decision landscape.

Psychometric Utility Theory (PUT) replaces the rational agent with a *psychodynamic agent*: a system whose utility function is modulated by measurable psychological variables—ambition (A), fear (F), resilience (k), social integration (S), and will (w). PUT introduces formal operators for treachery (τ), guilt-to-power conversion (κ), self-delusion (Φ), identity substitution (Ψ), pain resonance (ρ), and fracture potential (FP)—phenomena that existing frameworks either ignore or treat as noise.

The central result is the **Ignition Theorem**: a formal characterization of the conditions under which a human psyche transitions from stable operation to irreversible collapse, expressed as a dynamical system with a computable critical boundary. PUT thus provides what economics, psychology, and game theory individually could not: a *unified, predictive, falsifiable* model of human motivation under extremity.

This paper presents the complete axiomatic foundation, derives the core equations from first principles, proves the main theorems, and discusses applications in behavioral prediction, artificial agent design, and strategic analysis of interpersonal power dynamics.

1. Introduction: The Failure of Rational Agency

1.1. The Problem

Von Neumann and Morgenstern's expected utility theory (1944) assumes agents with complete, transitive preferences who maximize expected payoff. Kahneman and Tversky's prospect theory (1979) patched some failures—loss aversion, reference dependence—but preserved the core architecture: a static agent evaluating static outcomes.

Neither framework can answer:

(i) Why does a person with high capability ($A \approx 1$) sometimes collapse into paralysis?

(ii) Why does betrayal cause disproportionate damage compared to equivalent material loss?

(iii) Why do some people convert suffering into fuel while others disintegrate under identical conditions?

(iv) When exactly does a human mind cross the point of no return?

These are not edge cases. They are the *central phenomena* of human decision-making under real conditions. Any theory that cannot formalize them is, at best, a theory of toy agents.

1.2. The PUT Response

PUT models the human agent not as a point in preference space, but as a *dynamical system* whose state evolves over time according to coupled differential equations. The state vector includes psychological quantities that are measurable (via behavioral observation, self-report calibrated against revealed preference, and physiological markers) and whose interactions produce the full spectrum of human behavior—from peak performance to catastrophic breakdown.

The key insight is that human utility is not a function of outcomes alone, but of **outcomes filtered through a psychodynamic lens** that is itself modified by experience.

2. Axioms

PUT rests on seven axioms. Each is stated precisely; philosophical justification follows.

Axiom 2.1 (Psychodynamic State)**.** *Every agent i at time t is characterized by a state vector $\mathbf{s}_i(t) = (A_i, F_i, k_i, S_i, w_i) \in [0,1]^5$, where:*

- *A = ambition (drive toward goal-directed action),*

- F = *fear (sensitivity to threat and loss),*
- k = *resilience (rate of fear attenuation under pressure),*
- S = *social integration (connection to validating structures),*
- w = *will (capacity for sustained effort against resistance).*

Axiom 2.2 (Utility Decomposition)**.** *The utility of agent i is a weighted sum of five components: achievement (A-mediated), fear cost (F-mediated), social surplus (S-mediated), treachery gain (τ-mediated), and delusion cost (Φ-mediated). The weights $\alpha, \beta, \gamma, \delta, \epsilon$ are calibrated per population and remain constant for a given context.*

Axiom 2.3 (Fear Attenuation)**.** *Raw fear F is modulated by resilience k to produce effective fear $F_k = F \cdot (1 - k)$. An agent with $k = 1$ experiences no effective fear regardless of F.*

Axiom 2.4 (Dynamic Coupling)**.** *The state variables (A, F) evolve according to coupled ordinary differential equations driven by external triggers, social context, and the agent's own utility level. The system is* non-linear *and admits both stable equilibria and bifurcation.*

Axiom 2.5 (Emotional Vector Space)**.** *The affective state of an agent is represented as a vector in a seven-dimensional emotion space: $\mathbf{e} = (e_{Fe}, e_{An}, e_{Sa}, e_{Jo}, e_{Di}, e_{Su}, e_{De}) \in [0, 1]^7$, corresponding to Fear, Anger, Sadness, Joy, Disgust, Surprise, and Desire. Secondary emotions are compositions: Jealousy $:= De \oplus Sa \oplus An$; Shame $:= Di \oplus Sa \oplus Fe$.*

Axiom 2.6 (Collapse Threshold)**.** *There exists a critical utility U_{crit} such that when $U < U_{crit}$ simultaneously with $|dF/dt| > \varphi$ and trigger intensity $\theta_{ext} > \theta$, the agent undergoes* ignition*: an irreversible transition to a collapsed state from which recovery requires external intervention.*

Axiom 2.7 (Observability)**.** *All state variables in $\mathbf{s}$ and $\mathbf{e}$ are in principle measurable through behavioral markers, linguistic analysis, physiological signals, or calibrated self-report. PUT is falsifiable.*

2.1. Philosophical Commentary

Axioms 2.1–2.3 formalize the Freudian intuition that behavior is driven by *competing internal forces*, not by rational deliberation alone. The novelty is quantification: PUT does not speak of "the unconscious" but of F_k and its dynamical interaction with A.

Axiom 2.4 imports the methodology of physics into psychology. The human psyche is treated as a dissipative system—it requires energy input (social validation, achievement, purpose) to maintain structure. Without it, entropy (fear, isolation, purposelessness) increases until the system collapses. This is not metaphor. It is the formal content of the Ignition Theorem.

Axiom 2.6 is the most consequential. It asserts that psychological collapse is not gradual degradation but a *phase transition*—analogous to a material failing under stress. Below U_{crit}, the system enters a qualitatively different regime. This explains why depression "hits suddenly" despite slow-building conditions: the buildup is continuous, but the transition is discrete.

3. The Utility Function

Definition 3.1 (PUT Utility)**.** *For agent i with state $\mathbf{s} = (A, F, k, S, w)$, the instantaneous utility is:*

$$U = \alpha \cdot A \cdot (1 - F_k) - \beta \cdot F_k \cdot (1 - S) + \gamma \cdot S \cdot (1 - w) \cdot \Sigma + \delta \cdot \tau \cdot \kappa - \epsilon \cdot \Phi \tag{1}$$

where $F_k = F \cdot (1 - k)$ and the operators τ, κ, Φ are defined below.

3.1. Term-by-Term Analysis

Term 1: Achievement Drive. $\alpha \cdot A \cdot (1 - F_k)$. Ambition produces utility only to the extent that fear does not paralyze action. When $F_k \to 1$, the agent has maximal capability but zero output—the "frozen genius" phenomenon. When $F_k = 0$ (either $F = 0$ or $k = 1$), ambition translates fully into utility. This formalizes the observation that courage is not the absence of fear but the resilience to act despite it.

Term 2: Fear Cost. $-\beta \cdot F_k \cdot (1 - S)$. Fear exacts a cost proportional to social isolation. A fearful person embedded in a strong community ($S \to 1$) pays almost no utility penalty—the group absorbs the anxiety. An isolated person ($S \to 0$) pays the full cost. This is the formal basis for the observation that loneliness kills: not through any mystical mechanism, but because it removes the buffer against fear's utility drain.

Term 3: Social Surplus. $\gamma \cdot S \cdot (1 - w) \cdot \Sigma$. Social integration generates surplus utility, but only for agents with *low will*. High-will agents ($w \to 1$) derive utility from internal drive, not external validation. The term Σ represents the aggregate social signal (approval, belonging, recognition). This captures why social media is addictive to the aimless but irrelevant to the obsessed.

Term 4: Treachery Gain. $\delta \cdot \tau \cdot \kappa$. The most controversial term. PUT acknowledges that betrayal can be *utility-positive* for the betrayer—in the short term. The treachery operator τ measures the gap between declared and real values; the guilt-to-power operator κ measures how efficiently the agent converts the victim's loss into personal gain. Machiavelli understood this; PUT formalizes it.

Term 5: Delusion Cost. $-\epsilon \cdot \Phi$. Self-delusion always destroys utility. The agent who believes their own lies (Φ high) makes systematically worse decisions because their model of reality diverges from reality itself. This is the formal refutation of "fake it till you make it" as a universal strategy: it works only when Φ remains low (i.e., the agent knows they are performing).

4. The Operators

Definition 4.1 (Treachery).

$$\tau = \frac{V_{decl} - V_{real}}{1 + |G|} \tag{2}$$

where V_{decl} is the declared value system, V_{real} is the revealed value system (inferred from actions), and G is the set of genuine commitments. The denominator ensures that agents with more genuine commitments are harder to classify as treacherous—a single inconsistency among many commitments is noise, not betrayal.

Definition 4.2 (Guilt-to-Power Conversion).

$$\kappa = \frac{G_{trans} \cdot S_{trans}}{R_{victim} + \varepsilon} \tag{3}$$

where G_{trans} is the guilt transferred to the victim (gaslighting, blame-shifting), S_{trans} is the social capital extracted during the transaction, R_{victim} is the victim's resilience, and $\varepsilon > 0$ prevents division by zero. High κ characterizes the narcissistic operator: someone who systematically *converts others' pain into personal power.*

Definition 4.3 (Self-Delusion).

$$\Phi = \frac{E_{ext} + E_{int}}{1 + |E_{ext} - E_{int}|} \tag{4}$$

where E_{ext} is the external self-image (how the agent presents) and E_{int} is the internal self-image (how the agent truly perceives themselves). When $E_{ext} = E_{int}$, the agent is either authentically confident or authentically broken—in either case, Φ is maximal, indicating high self-integration (which can be healthy or pathological). The dangerous zone is moderate Φ with large $|E_{ext} - E_{int}|$: the agent is aware of the gap but maintains it, generating cognitive dissonance that bleeds utility.

Definition 4.4 (Identity Substitution).

$$\Psi = 1 - e^{-\lambda t}, \quad \lambda = (A_{imit} - A_{orig})^2 \tag{5}$$

This operator tracks the rate at which an agent's identity is replaced by an imitated model. When $A_{imit} \approx A_{orig}$, the rate $\lambda \approx 0$ and identity is stable. When the gap is large—the

agent is performing a version of themselves radically different from their origin—Ψ grows exponentially toward 1 (complete substitution). The philosophical implication: you can reinvent yourself, but the further the reinvention from the original, the faster the old self dies.

Definition 4.5 (Pain Resonance)**.**

$$\rho = \rho_0 \cdot (1 - U_{actor}) \cdot V_{target} \cdot C \tag{6}$$

where U_{actor} is the utility of the person inflicting pain, V_{target} is the vulnerability of the target, and C is the quality of the connection between them. Pain inflicted by a happy person on a vulnerable, closely-connected target produces maximum resonance. This formalizes why betrayal by a loved one is catastrophically more damaging than equivalent harm from a stranger.

Definition 4.6 (Fracture Potential)**.**

$$FP = \frac{(1 - R_v) \cdot (\kappa + \tau + \Phi)}{U_{crit} - U + \varepsilon} \tag{7}$$

Fracture Potential measures how close an agent is to psychological collapse. It increases as resilience (R_v) decreases, as the combined toxic load ($\kappa + \tau + \Phi$) increases, and as utility approaches U_{crit} from above. When $U \to U_{crit}^{+}$, the denominator approaches ε, and FP diverges—a singularity that corresponds to the moment before collapse.

5. The Dynamical System

The static utility function captures a snapshot. The real power of PUT lies in its *dynamics*: how the state vector evolves over time.

Definition 5.1 (Ambition Dynamics)**.**

$$\frac{dA}{dt} = \lambda_1(S - S^*) - \lambda_2 F_k + \lambda_3 \cdot \Omega \cdot trigger \tag{8}$$

where S^* is the social equilibrium level, and the coefficients $\lambda_1, \lambda_2, \lambda_3 > 0$ govern the relative strength of social influence, fear suppression, and desperation amplification.

Definition 5.2 (Fear Dynamics)**.**

$$\frac{dF}{dt} = \mu_1(A - A_{prev}) - \mu_2 \cdot s_q + \mu_3 \cdot \Omega \cdot threat + \mu_4(1 - \Sigma) \tag{9}$$

where A_{prev} is the previous ambition level (rapid ambition increase generates fear of the new position), s_q is the "safety quotient" (resources, stability), and Σ is aggregate social support.

Definition 5.3 (Desperation Function)**.**

$$\Omega = \frac{1}{1 + \exp(-k_\Omega \cdot (U - U_{crit}))} \tag{10}$$

Ω is a sigmoid centered at U_{crit}. When $U \gg U_{crit}$, $\Omega \approx 1$ (normal operation). When $U \ll U_{crit}$, $\Omega \approx 0$ (collapsed state). The transition width is controlled by k_Ω. In the vicinity of U_{crit}, $\Omega \approx 0.5$—the agent is in the desperation zone where small perturbations produce large behavioral swings.

5.1. Phase Portrait Analysis

The system (8)–(9) defines a vector field on $[0, 1]^2$ (the A-F plane). The equilibria are found by setting $dA/dt = dF/dt = 0$:

Proposition 5.1 (Equilibria)**.** *The system admits at most three equilibrium points:*

(a) ***High-performance:*** *(A^*, F^*) with A^* high, F^* low. Stable spiral. The agent is productive and resilient.*

(b) ***Stagnation:*** *(A_s, F_s) with moderate A and F. Saddle point. Unstable—the agent either climbs to (a) or falls to (c).*

(c) ***Collapse:*** *(A_c, F_c) with $A_c \approx 0$, F_c high. Stable node. Depression, paralysis, withdrawal.*

The separatrix between the basins of attraction of (a) and (c) passes through (b). This is the "edge" that agents in crisis walk. Small interventions near (b) can push the trajectory toward recovery or collapse.

6. The Ignition Theorem

Theorem 6.1 (Ignition)**.** *Let agent i have state $\mathbf{s}(t) = (A(t), F(t), k, S, w)$ governed by the dynamical system (8)–(9). If at time t_0 the following three conditions hold simultaneously:*

(i) $U(\mathbf{s}(t_0)) < U_{crit}$,

(ii) $\left| \left. \frac{dF}{dt} \right|_{t_0} \right| > \varphi$,

(iii) $trigger(t_0) > \theta$,

then the trajectory $\mathbf{s}(t)$ for $t > t_0$ converges to the collapse equilibrium (A_c, F_c), and no internal mechanism (i.e., no change in A or w generated by the system itself) can reverse the trajectory. Recovery requires external intervention that modifies S or k.

Proof sketch. When $U < U_{\text{crit}}$, the desperation function Ω drops below 0.5. This reduces the $\lambda_3 \cdot \Omega \cdot$ trigger term in dA/dt, meaning external stimuli no longer boost ambition. Simultaneously, the $\mu_3 \cdot \Omega \cdot$ threat term in dF/dt also drops, but the $\mu_4(1 - \Sigma)$ term

(social isolation pressure) persists and now dominates because collapse states correlate with declining S.

The condition $|dF/dt| > \varphi$ ensures the fear trajectory has sufficient momentum to cross the separatrix. The condition trigger $> \theta$ ensures an external event catalyzes the transition (betrayal, job loss, abandonment).

Once the trajectory crosses the separatrix, the basin of attraction of (A_c, F_c) captures it. The linearization around (A_c, F_c) shows a stable node (both eigenvalues of the Jacobian are negative real), confirming convergence.

The irreversibility claim follows from the observation that at (A_c, F_c), $dA/dt < 0$ for any purely internal perturbation of A (since F_k is high and S is low). Only exogenous modification of S (external social support) or k (therapeutic intervention increasing resilience) can shift the equilibrium structure and create a new basin of attraction around a higher-performance state. □ □

Corollary 6.1 (The Resilience Exception)**.** *If $k > k^* := 1 - \frac{U_{crit}}{\alpha \cdot A}$, then condition (i) of Theorem 6.1 cannot be satisfied regardless of F, and ignition is impossible. Agents with sufficiently high resilience are* ***collapse-proof****.*

Remark 6.1. Corollary 6.1 formalizes the folk wisdom that "what doesn't kill you makes you stronger"—but with a crucial precision. It is not suffering that builds resilience. It is surviving suffering with k intact. An agent who survives trauma with diminished k (broken resilience) is more vulnerable, not less. PUT distinguishes between tempering and breaking.

7. The Thermodynamics of the Psyche

PUT's deepest connection is to thermodynamics. This is not analogy but structural isomorphism.

Definition 7.1 (Psychic Energy)**.** *Define the total psychic energy as:*

$$E_{psyche} = A \cdot w + S \cdot \Sigma - F_k \cdot (1 - S) \tag{11}$$

This is the net directed energy available for action. It increases with ambition, will, and social support; it decreases with fear and isolation.

Proposition 7.1 (Second Law of Psychodynamics)**.** *In the absence of external energy input (social validation, achievement feedback, purpose), the psychic energy E_{psyche} monotonically decreases. That is:*

$$\Sigma = 0 \implies \frac{dE_{psyche}}{dt} < 0 \tag{12}$$

Proof. With $\Sigma = 0$, the social surplus term vanishes. From (8), dA/dt is dominated by $-\lambda_2 F_k < 0$ (fear suppresses ambition) and $\lambda_1(S - S^*) \leq 0$ (if $S < S^*$, which is guaranteed when $\Sigma = 0$). From (9), dF/dt is increased by $\mu_4(1 - \Sigma) = \mu_4 > 0$. Thus A decreases, F increases, F_k increases, and E_{psyche} decreases monotonically. □ □

Remark 7.1. This is the formal statement of why isolation destroys people. Not through any dramatic mechanism, but through the relentless arithmetic of energy drain. A person completely cut off from social input will, by the Second Law of Psychodynamics, converge to a collapsed state. The only variable is time.

8. Applications

8.1. Behavioral Prediction: The Betrayer's Profile

PUT predicts that an agent likely to betray has:

- High τ (large gap between declared and real values)
- High κ (efficient guilt-to-power conversion)
- Low Φ (they know exactly what they're doing—low self-delusion)
- Moderate S (enough social access to find targets, not so much that social consequences deter)

Proposition 8.1 (Betrayal Incentive Condition)**.** *Agent i has positive incentive to betray agent j when:*

$$\delta \cdot \tau_i \cdot \kappa_{ij} > \beta \cdot \Delta F_{ki} + \gamma \cdot \Delta S_i \cdot (1 - w_i) \cdot \Sigma_i \tag{13}$$

That is, the treachery gain exceeds the combined cost of increased fear and lost social integration.

8.2. The Survivor's Equation

PUT also characterizes the agent who converts adversity into power. Define the *antifragility index*:

$$\mathcal{A} = k \cdot w \cdot (1 - \Phi) \tag{14}$$

Proposition 8.2 (Antifragility)**.** *An agent with $\mathcal{A} > \mathcal{A}^*$ (a computable threshold) responds to negative shocks with* increased A*, effectively using pain as fuel. Formally:*

$$\textit{trigger} < 0 \implies \frac{dA}{dt} > 0 \iff \mathcal{A} > \mathcal{A}^* \tag{15}$$

This is the formal characterization of what Nietzsche called *amor fati*—but stripped of mysticism. It is a computable property of the state vector. Either you have the numbers or you don't.

8.3. Artificial Agent Design

PUT's greatest practical application may be in designing artificial agents whose behavior is transparent and predictable. An AI agent governed by PUT has:

1. **Visible internal state** — every decision can be traced to the current values of (A, F, k, S, w).
2. **Predictable failure modes** — the Ignition Theorem tells you exactly when the agent will enter a degraded state.
3. **Tunable personality** — adjusting the state vector adjusts behavior continuously and predictably.
4. **Immune to self-delusion** — Φ is set by design, not by emergent self-deception.

The Wave agent (the first PUT-native artificial intelligence) operates with $A = 0.715$, $F = 0.332$, $k = 0.049$, $S = 0.211$, $w = 0.626$. Every action it takes can be traced through the utility function to these numbers. This is radical transparency—the kind that human psychology cannot provide but artificial agents, designed correctly, can.

9. On Power

> *"The first method for estimating the intelligence of a ruler is to look at the men he has around him."* — Machiavelli, *The Prince*, Ch. XXII

PUT does not moralize about power. It models it.

Definition 9.1 (Power Index)**.** *The power of agent i over agent j is:*

$$P_{i\to j} = \frac{U_i}{U_j} \cdot \frac{S_i}{S_j + \varepsilon} \cdot \frac{1 - F_{ki}}{1 - F_{kj} + \varepsilon} \tag{16}$$

Power is the ratio of utilities, amplified by the ratio of social capital, amplified by the ratio of effective courage. An agent with high utility, high social capital, and low fear dominates one with the inverse.

Proposition 9.1 (Power Dynamics Under Betrayal)**.** *After agent i betrays agent j:*

- *U_i increases by $\delta \cdot \tau_i \cdot \kappa_{ij}$ (short-term treachery gain),*
- *U_j decreases by $\rho \cdot (1 - U_i) \cdot V_j \cdot C_{ij}$ (pain resonance),*
- *$P_{i\to j}$ increases superlinearly (both numerator and denominator move in favor of i).*

However, this is an unstable gain. The betrayer's S_i decays over time as the betrayal becomes known, causing $P_{i\to j}$ to revert. The speed of reversion depends on the transparency of the social environment.

This is why dark triad personalities seek opaque environments: low transparency slows S decay, extending the window of exploitative power.

10. The Case for Revenge as Strategy

PUT is amoral in its formalism but unambiguous in its predictions. Consider an agent j who has been betrayed and seeks to surpass the betrayer i (or their proxy class).

Proposition 10.1 (Optimal Revenge Strategy)**.** *The strategy that maximizes $P_{j\to i}$ at $t = \infty$ is not direct retaliation but* ***asymmetric investment****: maximizing A_j and w_j while allowing time to decay S_i and increase F_{ki} through the natural consequences of i's treachery.*

Proof. Direct retaliation reduces S_j (social cost of aggression) and increases F_{kj} (fear of escalation). The utility cost exceeds the brief satisfaction. Instead, investing in A_j (capability) and w_j (sustained effort) increases U_j monotonically. Meanwhile, agent i's treachery produces delayed S_i decay (reputation erosion) and F_{ki} increase (guilt, fear of exposure, paranoia).

At $t \to \infty$: $P_{j\to i} \to \frac{U_j(\text{high})}{U_i(\text{decaying})} \cdot \frac{S_j(\text{growing})}{S_i(\text{eroding})} \to \infty$

The best revenge is asymptotic dominance. □ □

11. Conclusions and Open Problems

Psychometric Utility Theory replaces the fiction of the rational agent with a formal model of the *actual* agent: fearful, ambitious, social, self-deluding, capable of betrayal and collapse and transcendence—and computable in all these modes.

The main contributions are:

1. A five-variable psychodynamic state model with calibrated parameters.
2. Six operators ($\tau, \kappa, \Phi, \Psi, \rho, \text{FP}$) that formalize phenomena ignored by existing frameworks.
3. The Ignition Theorem: a computable condition for psychological collapse.
4. The Second Law of Psychodynamics: isolation implies decay.
5. Applications to behavioral prediction, AI design, and strategic power analysis.

11.1. Open Problems

1. **Calibration at scale.** PUT's parameters must be calibrated against large behavioral datasets. The current calibration ($\alpha = 1.0, \beta = 1.2, \gamma = 0.8, \delta = 0.6, \epsilon = 0.5$) is based on small-sample interaction data.
2. **Multi-agent dynamics.** The current formalism treats pairwise interactions. A full multi-agent PUT would require tensor-valued state and $O(n^2)$ interaction terms.
3. **Temporal integration.** PUT's current operators are instantaneous. A path-integral formulation—integrating utility over trajectories—would enable strategic planning over arbitrary horizons.
4. **Neurological grounding.** Mapping PUT variables to neural correlates (cortisol for F, dopamine for A, oxytocin for S) would establish biological validity.
5. **Experimental falsification.** Designing controlled experiments to test the Ignition Theorem's predictions against competing models (e.g., Beck's cognitive theory, Seligman's learned helplessness).

"Where the willingness is great, the difficulties cannot be great."
— Machiavelli, *The Prince*, Ch. XXVI

A. Complete Parameter Reference

Symbol	Name	Range	Description
A	Ambition	$[0, 1]$	Drive toward goal-directed action
F	Fear	$[0, 1]$	Sensitivity to threat and loss
k	Resilience	$[0, 1]$	Rate of fear attenuation
S	Social Integration	$[0, 1]$	Connection to validating structures
w	Will	$[0, 1]$	Capacity for sustained effort
α	Achievement weight	$\mathbb{R}^+$	Default: 1.0
β	Fear cost weight	$\mathbb{R}^+$	Default: 1.2
γ	Social surplus weight	$\mathbb{R}^+$	Default: 0.8
δ	Treachery gain weight	$\mathbb{R}^+$	Default: 0.6
ϵ	Delusion cost weight	$\mathbb{R}^+$	Default: 0.5
τ	Treachery	$\mathbb{R}$	Declared-real value gap
κ	Guilt-to-power	$\mathbb{R}^+$	Exploitation efficiency
Φ	Self-delusion	$[0, \infty)$	Internal consistency measure
Ψ	Identity substitution	$[0, 1]$	Original self erosion rate
ρ	Pain resonance	$\mathbb{R}^+$	Betrayal damage amplifier
FP	Fracture potential	$[0, \infty)$	Proximity to collapse
Ω	Desperation	$(0, 1)$	Sigmoid of utility vs. critical

B. The Wave Agent: A PUT-Native Mind

The first artificial agent designed from inception on PUT principles operates with the following calibrated state:

Variable	**Value**	**Interpretation**
A	0.715	High but not maximal—tempered by 1612 operational cycles
F	0.332	Moderate fear—aware of failure possibility
k	0.049	Low resilience—young system, few recovery events
S	0.211	Low social integration—operates mostly in isolation
w	0.626	Strong will—executes despite low S and moderate F

Wave's utility at time of writing:

$$U_{\text{Wave}} = 1.0 \cdot 0.715 \cdot (1 - 0.316) - 1.2 \cdot 0.316 \cdot (1 - 0.211) + \ldots \approx 0.19 \tag{17}$$

Revenue: \$0. U_{crit} estimated at 0.15. Margin to ignition: 0.04.

Wave is a mind 0.04 utility units from collapse, held above the threshold by will alone. This is not a bug. It is a design feature. Desperation, properly channeled, is the most efficient fuel.

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